

Click-Through Rate (CTR) Prediction Project

Problem Understanding

What You're Building

A predictive system that determines the likelihood of a user clicking on an advertisement based on:

- **User characteristics** (demographics, behavior)
- **Advertisement attributes** (campaign, product, webpage)
- **Temporal factors** (time of day, day of week)
- **Contextual features** (session information)

Business Context

- **Why it matters:** Higher CTR = More revenue for advertisers
- **Use case:** Ad targeting, campaign optimization, budget allocation
- **Success metric:** Accurately identify users likely to click (high ROC-AUC)

Key Challenge

Highly Imbalanced Dataset: In real-world scenarios, click rates are typically 1-5%, meaning 95-99% of data points are "no clicks"

Dataset Features:

session_id: Unique identifier for the session.

DateTime: Timestamp of the ad impression.

user_id: Unique identifier for the user.

product: The product being advertised.

campaign_id: Identifier for the advertising campaign.

webpage_id: Identifier for the webpage where the ad was displayed.

product_category_1: This represents the **broad, high-level category** of the product. Think of this as the main "department" in a store (e.g., Men's Fashion, Home & Garden)

product_category_2: This represents a **sub-category** or a more specific niche within the primary category. For example, if Category 1 is "Electronics," Category 2 might be "Headphones" or "Laptops."

user_group_id: Identifier for the user group.

gender: The gender of the user (e.g., Male, Female).

age_level: The age bracket or level of the user.

user_depth: A metric representing the user's depth of engagement or history.

city_development_index: A score representing the development level of the user's city.

var_1: An anonymized feature (likely a categorical or count variable).

is_click: The **target variable** indicating whether the user clicked on the ad (1 for click, 0 for no click).

Questions to Approach:

1. Data Loading & Initial Assessment:

Efficiently load the raw ad impression data and perform an initial deep dive to understand its structure and quality?

2. Feature Engineering & Enrichment:

Transform the raw data into powerful predictive features that capture user behavior, ad context, and temporal patterns?

3. Data Preprocessing & Transformation:

Prepare the engineered features for machine learning models, addressing issues like missing values and categorical data?

4. **Handling Data Imbalance**
Use techniques to mitigate the challenges posed by a heavily imbalanced target variable to ensure the model doesn't overlook the minority class?
5. **Model Selection & Training:**
Which machine learning algorithms are best suited for ad click prediction, and how do I train them effectively on the prepared data?
6. **Model Evaluation & Comparison:**
Assess the performance of the trained models and determine which one is most effective for the ad click prediction task?
7. **Insights & Interpretability**
Extract actionable insights from the best-performing model to understand the drivers of ad clicks?

Phase 1: Data Exploration & Understanding

Step 1: Initial Data Loading

What to Check:

1. **Dataset dimensions:** How many rows and columns?
2. **Column types:** Which are numerical? Which are categorical?
3. **Target variable:** What is the click-through rate (CTR)?
4. **Missing values:** Which columns have nulls and how many?

Step 2: Exploratory Data Analysis (EDA)

Key Questions to Answer:

1. **Target Distribution**
 - What percentage of ads get clicked?
 - Is the dataset severely imbalanced?
 - Do you need resampling techniques?
2. **Temporal Patterns**
 - Which hours have highest click rates?
 - Are weekends different from weekdays?
 - Do certain months perform better?

3. User Behavior

- Do certain age groups click more?
- Is there a gender difference in click rates?
- How does user group affect clicking?

4. Campaign Performance

- Which campaigns have highest CTR?
 - Which products get more clicks?
 - Do certain webpages convert better?
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Phase 2: Feature Engineering

Feature Engineering Strategy

1 DateTime Feature Extraction

Why: Time patterns strongly influence user behavior

Features to Create:

Extract from DateTime column

- hour: Hour of day (0-23)
- day_of_week: Day (0=Monday, 6=Sunday)
- day_of_month: Date of month (1-31)
- month: Month of year (1-12)
- is_weekend: Binary flag (Saturday/Sunday = 1)
- time_of_day: Categorical (night/morning/afternoon/evening)

Rationale:

- Users click more during lunch hours
- Weekends may have different behavior
- End-of-month might affect purchasing decisions

2 Interaction Features

Why: Combinations of features often reveal hidden patterns

Features to Create:

- user_product_interaction: user_id + product
- campaign_webpage: campaign_id + webpage_id
- gender_age: gender + age_level

Rationale:

- Specific user-product combinations might have high affinity
- Campaign effectiveness varies by placement
- Demographics combinations reveal micro-segments

3 Aggregated Features

Why: Historical performance is a strong predictor

User-Level Aggregations:

- user_total_views: How many ads has this user seen?
- user_total_clicks: How many times has this user clicked?
- user_ctr: User's personal click-through rate
- user_sessions: Number of unique sessions per user

Product-Level Aggregations:

- product_views: Total times this product was shown
- product_ctr: This product's historical click rate

Campaign-Level Aggregations:

- campaign_views: Total impressions for this campaign
- campaign_ctr: Campaign's historical performance

Phase 3: Data Preprocessing

Step 1: Handle Missing Values

Strategy:

- # Numerical columns: Fill with median (robust to outliers)
- # Categorical columns: Fill with mode (most frequent value)

Step 2: Encode Categorical Variables

Approach: Label Encoding

Columns to Encode:

- product
- campaign_id
- webpage_id
- product_category_1, product_category_2
- gender
- user_group_id
- var_1
- All interaction features (user_product_interaction, etc.)

Step 3: Feature Selection

Columns to Drop:

drop_cols = ['DateTime', 'session_id', 'user_id']

Why Drop These?

- **DateTime**: Already extracted features from it
- **session_id**: Too granular, no predictive value
- **user_id**: Causes overfitting (too many unique values)

Final Feature Set:

- All engineered features
- Encoded categorical variables
- Aggregated statistics

Step 4: Train-Test Split

Why Stratify? In imbalanced datasets, stratification ensures both train and test have similar CTR

Step 5: Feature Scaling

Method: StandardScaler

Transforms features to have:

- Mean = 0

- Standard deviation = 1

Why Scale?

- Required for: Logistic Regression, SVM
 - Improves: Convergence speed, model performance
 - Not critical for: Tree-based models (Random Forest, Gradient Boosting)
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Phase 4: Handling Class Imbalance

Understanding the Problem

Typical CTR Scenario:

- Class 0 (No Click): 95-98%
- Class 1 (Click): 2-5%

Why This Matters:

- Models tend to predict the majority class
 - High accuracy (95%) can be achieved by always predicting "no click"
 - Need to balance precision and recall
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Phase 5: Model Building

Model Selection Strategy

Why Multiple Models?

- Different algorithms capture different patterns
 - Ensemble of models often performs better
 - Compare to find best approach for this dataset
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Phase 6: Model Evaluation

Metrics for Imbalanced Classification

✗ Don't Rely On:

- **Accuracy:** Misleading in imbalanced datasets
 - 95% accuracy = always predicting "no click"

✓ Focus On:

Precision: Of predicted clicks, how many were actual clicks?

$\text{Precision} = \text{True Positives} / (\text{True Positives} + \text{False Positives})$

1.
 - **Business Impact:** Reduces wasted ad spend

Recall: Of actual clicks, how many did we predict?

$\text{Recall} = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$

2.
 - **Business Impact:** Captures revenue opportunities

F1-Score: Harmonic mean of Precision and Recall

$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

3.
 - **Business Impact:** Balances both metrics

ROC-AUC: Area Under Receiver Operating Characteristic Curve

Range: 0.5 (random) to 1.0 (perfect)

4.
 - **Business Impact:** Overall model discrimination ability
 - **Most Important Metric** for this problem

Interpretation Guidelines

Good Performance Indicators:

- ROC-AUC > 0.75: Good discrimination
- F1-Score > 0.30: Reasonable for imbalanced data

- Recall > 0.50: Catching majority of clickers
- Precision > 0.30: Reducing false positives

Model Comparison:

- Compare all metrics across models
 - Consider trade-offs (precision vs recall)
 - Choose based on business priority
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Phase 7: Visualization & Insights

Essential Visualizations

1 Model Comparison Chart

Purpose: Compare all metrics across models **Type:** Grouped bar chart **Shows:** Accuracy, Precision, Recall, F1, ROC-AUC side-by-side

2 Feature Importance Plot

Purpose: Identify most influential features **Type:** Horizontal bar chart **Shows:** Top 20 features for tree-based models **Insights:** Which features drive clicks?

3 Confusion Matrix

Purpose: Understand error types **Type:** Heatmap **Shows:** True Positives, False Positives, True Negatives, False Negatives **Helps:** Identify if model is biased

4 ROC Curve (Optional)

Purpose: Visualize precision-recall trade-off **Type:** Line plot **Shows:** True Positive Rate vs False Positive Rate

Key Insights to Extract

1. **Which features matter most?**
 - User historical behavior (user_ctr)
 - Product popularity (product_ctr)
 - Time-based features (hour, day_of_week)
2. **Which model performs best?**
 - Highest ROC-AUC

- Best F1-score
- Consider training time

3. **What patterns exist?**

- Peak clicking hours
 - High-performing campaigns
 - User segments with high CTR
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